A Computational Study of Jet Observables in ep Collisions at $\sqrt{s} = 140$ GeV

Antonio Perillo

Advisor: Sevil Salur

Abstract

With the Electron-Ion Collider (EIC) anticipated to be operational within the next ten years, simulation studies of the physics to be investigated at the EIC have largely been at the forefront of modern nuclear physics research. As a part of this effort, this paper presents a computational study of ep collisions at center-of-mass energies of 140 GeV, representative of the future EIC beam parameters. Using PYTHIA, we generate 10^6 events, then apply the anti-k_t jet clustering algorithm provided in FastJet to reconstruct the jets created in these collisions. After validating our kinematic distributions, we tag the jets as either quark- or gluon-initiated. We then examine the utility of generalized angularity functions (GAFs) with various weights to systematically differentiate between quark and gluon jets. Ultimately, from this methodology, this thesis was unable to make any definitive conclusions regarding the distinguishability of quark and gluon jet shapes using the GAFs.

Keywords: Quantum Chromodynamics, Jets, Electron-Ion Collider

1 Introduction and Theory

1.1 The Standard Model

The most comprehensive theory of the particles and interactions that govern our observable universe is described within a mathematical compendium known as The Standard Model (SM). The SM is the culminating achievement of 20th-century physics; thus, no single paper could systematically unravel the theory in its entirety. Regardless, several key takeaways are foundational in the development of this thesis and therefore must be explicitly mentioned. Consider the following pictorial representation of the known particles of the SM in Figure 1:

The known particles can largely be distinguished into two categories: the matter particles or fermions (seen on the left), and the force carriers, or bosons (seen in orange). These particles fundamentally differ inasmuch as fermions possess half-integer spin (intrinsic angular momenta) and bosons possess integer spin. Within the fermions, there are yet subdivisions: quarks and leptons. Quarks are the building blocks of more complex particle structures known as hadrons. Hadrons are categorized as either baryons, which consist of three bound quarks, or mesons, which are highly unstable quark-antiquark pairs. The familiar nucleons, i.e., the proton and neutron, in a simplified picture, consist of two up and a down quark (uud), and two down and an up quark (ddu), respectively. The leptons consist of the electron, its two heavier counterparts, the muon and tau, as well as their corresponding neutrinos. These matter particles interact via mediating, or gauge, bosons. Each of these gauge bosons mediates a fundamental force of nature. The gluon mediates the strong nuclear force, the photon mediates the electromagnetic force, and the W and Z bosons mediate the weak nuclear force. When one considers the interactions of the known particles, particularly in high-energy environments, one must evaluate the system through the lens of the SM.

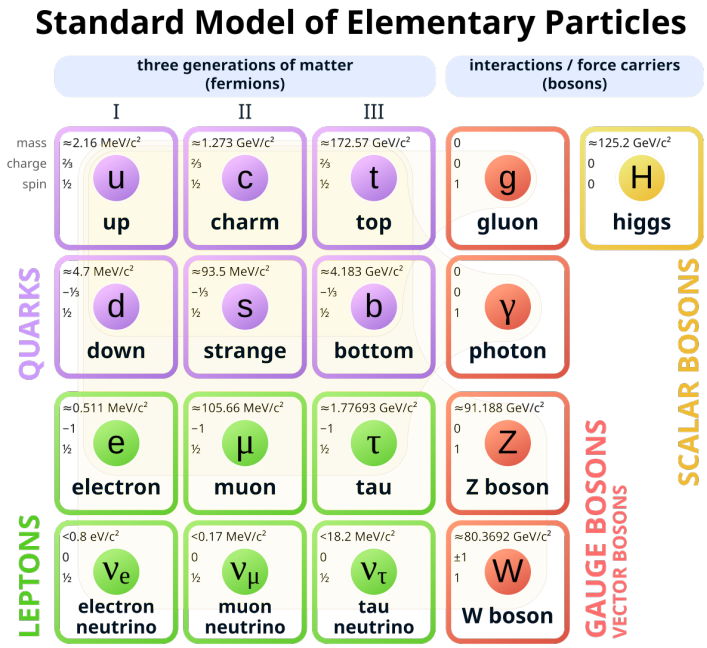


Figure 1: The Standard Model of Particle Physics [16].

1.2 Quantum Chromodynamics

Furthermore, this thesis is primarily concerned with the interactions within the nucleons. The physics of this domain is described by yet another substantial theory, known as ***Quantum Chromodynamics*** (QCD). QCD describes the interactions between quarks and gluons, the building blocks of the nucleons and the force carriers that hold them together, respectively. There are three generations of quarks (see Figure 1): the up and down, charm and strange, and top and bottom. Each quark carries a fundamental quantity analogous to electric charge, which physicists conveniently call color. Unlike electric charge, color is not intrinsically an observable quantity, but rather, quarks form bound states that are "colorless", a phenomenon referred to as *color confinement*. For example, the proton consists of a red, blue, and green quark, making it "white" or colorless. Similarly, π^+ is made of a quark and an anti-quark, or a quark with color and a quark with "anti-color", forming a colorless state. The necessity for color as a degree of freedom lies in the fact that quarks are fermions, meaning they obey the Pauli Exclusion Principle, which states that no two fermions can simultaneously occupy the same quantum state. If not for this additional degree of freedom, baryons would violate this fundamental principle of nature. Moreover, QCD is a rich scientific theory of much contemporary interest. In contrast to quantum electrodynamics (QED), the quantum description of the electric field, where the force carriers, photons, do not generally interact, in QCD, the gluons do interact, resulting in a complex emergence of mechanisms not found elsewhere in nature. In addition, calculations in QCD are often incredibly difficult due to its *non-perturbative* nature [11]. Traditionally, in quantum field theory, when one cannot analytically calculate the solution to a problem, one resorts to approximations via *perturbation theory.* Qualitatively, this involves expanding the problem in orders of the *coupling constant*, a dimensionless number that governs the strength of the interaction of concern. To perform this expansion, however, the coupling constant must be sufficiently small such that higher-order terms diminish. This is not generally the case in QCD due to its running coupling constant.

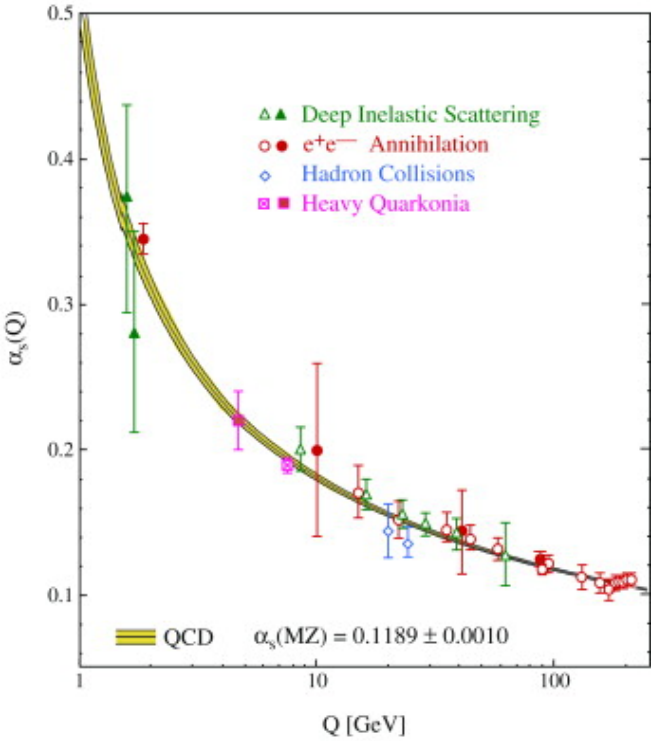


Figure 2: A plot demonstrating the running QCD coupling constant, α_S. The bold yellow line represents theoretical calculations of α_S, while the data points represent several empirical methods for measuring α_S as a function of momentum transfer. Evidently, at high momentum transfer, the coupling constant asymptotically approaches a small, finite value, allowing for the application of perturbation theory. However, at small momentum transfer, the coupling constant becomes unbounded, meaning the strength of the QCD interaction increases, which prohibits the use of perturbation theory. Ingenious workarounds, such as lattice QCD, allow for theoretical calculations at low momentum transfer, despite their non-perturbative nature [2].

From Figure 2, one can see that the coupling constant is not constant, but rather depends on the energy scales being considered. Consequently, perturbation theory cannot be applied to the low-energy regime. As a result, many outstanding problems in QCD require empirically measured parameters, making experimentation and simulation studies important for progress.

1.2.1 Particle Collisions and Their Probes

Particle collisions are one such experimental procedure for unraveling the nature of the nucleons and have been of considerable scientific interest for the last century. Specifically, particle collisions provide a cohesive bridge between theory and experiment, as accurately predicting the outcome of a scattering experiment requires robust knowledge of the target and the scatterer. Similarly, results from the scattering experiment can inform the theory describing the collision event. Perhaps the first, and certainly one of the most important, Rutherford, Geiger, and Marsden's seminal gold-foil experiment demonstrated that scattering experiments provide an empirically robust way for studying the structure of a scattering target. The fundamental result of their experiment was that the nucleus of the atom is composed of a dense concentration of positive charge, which we now regard as the protons [13]. As experimental capabilities improved over the next 60 years, physicists at the Stanford Linear Accelerator Center (SLAC), were able to perform similar studies, but at much higher energies. While Rutherford's α-particles had energies $\cong$ 5 MeV, *ep* collisions at SLAC accessed center-of-mass energies of up to 6 GeV.

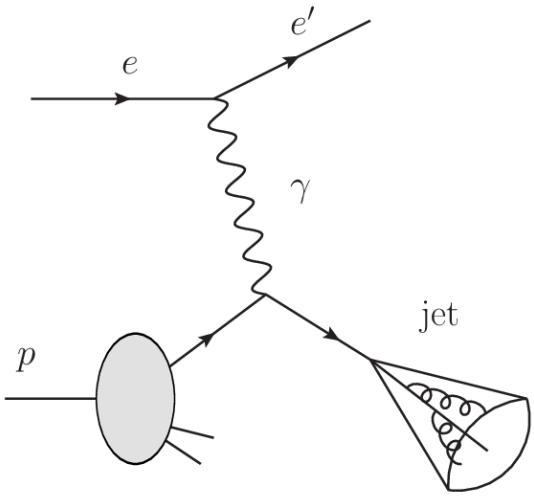


Figure 3: A Feynman diagram representing deep-inelastic scattering (DIS), the process in which a high-energy electron scatters off a quark inside the proton, resulting in fragmentation of the proton structure, which can be identified as the collimated stream of particles, that is, jets that are created, consequently [7].

.

At these energies, the electron does not simply scatter from the Coulomb potential of the proton, as it does at low energies, but rather, it scatters from the constituent quarks inside the proton, revealing a complex substructure. These deep-inelastic scattering (DIS) experiments at SLAC were the first to unveil the quark structure of the proton, further demonstrating the triumph of scattering experiments to study the underlying physics of particles [14]. Figure 3 provides a Feynman diagram representing the process of DIS. Namely, a high-energy electron, through the exchange of a highly virtual photon, scatters off a quark inside the proton. This quark, which is effectively "knocked out" of the proton, hadronizes, creating a collimated stream of particles, called a ***jet***, which can be experimentally detected. These jets evidently contain important information about the initial-scattered quark; thus, they act as our experimental proxy for probing the underlying QCD physics of the interaction.

1.2.2 The Kinematics of Particle Collisions

To characterize and analyze the jets created in these collisions, or more generally, to analyze the interactions between particles in a scattering experiment, one must have a well-defined methodology for extracting well-defined observables.

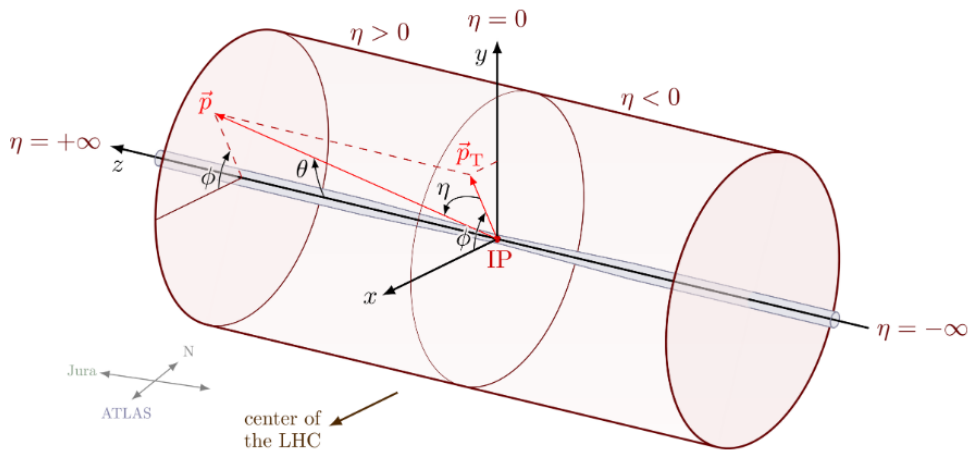


Figure 4: A drawing demonstrating a pertinent coordinate system for a particle collison experiment. The light pink cylindrical figure represents a detector with full azimuthal coverage. The thinner, gray cylinder represents a beamline where particles are directed to collide. [15].

Specifically, when diagnosing relativistic particle collisions, one must use the language of the Special Theory of Relativity (SR). This thesis will not provide a robust derivation of the consequences of SR, but rather, will introduce well-known consequences, as needed, for the practical development of understanding pertinent kinematic variables in particle collisions. To begin, SR postulates that physical phenomena (called *events*) are described by the 4-dimensional, spacetime, formally called Minkowski space. This demands four variables to describe any event in this spacetime, namely, (ct, x, y, z), where t represents time, c represents the speed of light, and (x, y, z) represent the Cartesian spatial coordinates. For notational simplicity, physicists typically refer to these coordinates as (x_0, x_1, x_2, x_3): we call these 4-vectors. SR provides a prescription for determining the relationship between events in coordinate systems that move with constant velocity with respect to each other, such that the length element in Minkowski space, denoted by:

$$ds^2 = -cdt^2 + dx^2 + dy^2 + dz^2 \tag{1}$$

is invariant regardless of the coordinate system. As a consequence, coordinates are mapped to other inertial systems through a *Lorentz Transformation*.

Let us now consider the case of particle collisions. Consider a particle moving with constant velocity v_z (towards the left in Figure 4). There are two pertinent inertial frames to consider: the laboratory frame and the particle's rest frame. We will denote the particle in the laboratory frame as x^ν. Transforming into the particle's frame, moving with velocity v_z, amounts to performing a Lorentz transformation on the laboratory 4-vector:

$$x^{\bar{\mu}} = \Lambda^{\bar{\mu}}_{\nu} x^\nu \tag{2}$$

where:

$$\Lambda^{\bar{\mu}}_{\nu} = \begin{pmatrix} \cosh y & 0 & 0 & -\sinh y \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sinh y & 0 & 0 & \cosh y \end{pmatrix} \tag{3}$$

where $y = \tanh^{-1} v$ is the *rapidity*. Because we are "boosting" along v_z, we can write:

$$y = \tanh^{-1} v_z \tag{4}$$

$$= \tanh^{-1} \frac{\gamma m v_z}{\gamma m} \tag{5}$$

$$= \tanh^{-1} \frac{p_z}{E} \tag{6}$$

$$= \frac{1}{2} \ln \frac{1 + p_z/E}{1 - pz/E} \tag{7}$$

$$= \frac{1}{2} \ln \frac{E + p_z}{E - p_z} \tag{8}$$

This can be shown to be related to the emission angle, θ, that subtends the beamline (see Figure 4), from which we can define the pseudorapidity, η:

$$\eta \equiv \frac{1}{2} \ln \frac{|\boldsymbol{p}| + p_z}{|\boldsymbol{p}| - p_z} = -\ln(\tan(\frac{\theta}{2})) \tag{9}$$

This, in combination with ϕ, which subtends the conical region circumscribing the beam, provides us with a complete description of the angular variables describing particles ejected from the beam axis. Because we are interested in the result of the particle collision, we utilize the transverse momentum, p_t, as our dynamic kinematic variable in contrast to $|\boldsymbol{p}|$, to avoid contamination from the momentum provided by the beamline itself. Therefore, with η, ϕ, and p_t, we have a systematic way of characterizing relativistic particle collisions that correspond to detector measurables.[1]

1.3 The Electron-Ion Collider

The ***Electron-Ion Collider*** (EIC) is an experimental modification to the current Relativistic Heavy-Ion Collider (RHIC) at Brookhaven National Laboratory (BNL) planned to be operational in the early 2030s. The EIC will feature highly polarized electron and proton beams and have ion beams of nuclei ranging from deuterium to uranium [1]. Additionally, the EIC will have variable center-of-mass energies ranging from 20-140 GeV, and be able to generate electron-nucleon luminosities of up to $10^{33} - 10^{34} \mathrm{cm}^{-2}\,\mathrm{s}^{-1}$ [1]. The fundamental difference between RHIC and the EIC is the species of the colliding particles. Whereas RHIC collides heavy ions, the EIC will instead collide electrons with protons and heavy ions.

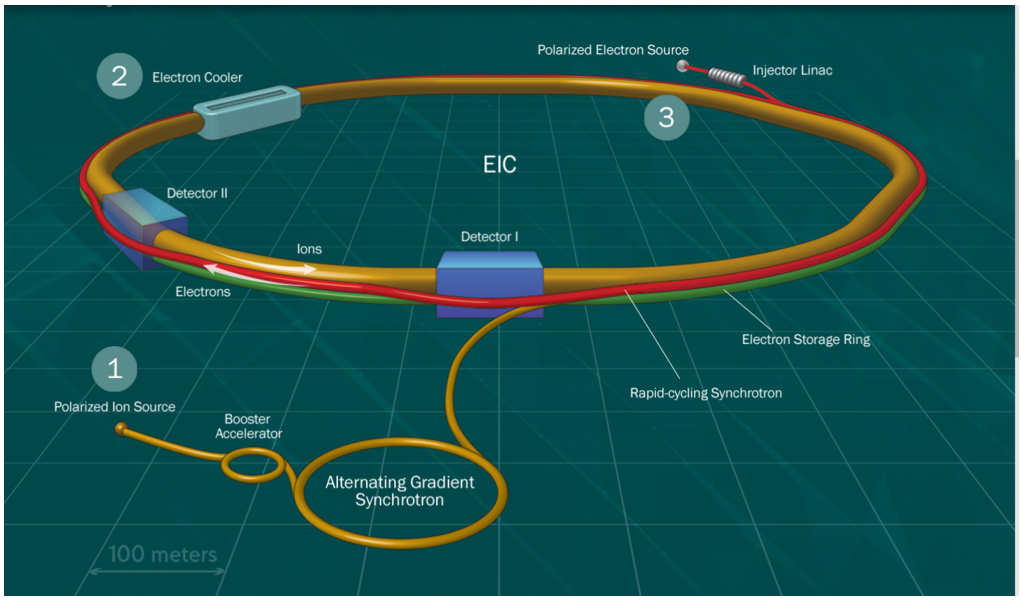


Figure 5: A tentative schematic for the future Electron-Ion Collider [9].

[1] This section was largely inspired by the work of Ravi Koka [11].

To understand why this seemingly subtle modification is significant, first consider the preliminary discussion regarding the SM that this thesis begins with. Recall that interactions between matter particles are mediated by the bosons. However, interactions only occur between matter particles that possess similar intrinsic quantities. For example, electromagnetic interactions only occur between particles that possess electric charge. Similarly, strong interactions only occur between particles that possess color charge. In heavy-ion collisions, hundreds of partons (quarks and gluons) interact, creating an ensemble of events that becomes cumbersome to deconstruct in order to extract meaningful information about any particular interaction. The electron, however, does not interact strongly, meaning these "noisy" interactions are not pertinent. Effectively, the electron scatters directly off a quark, providing precise measurements of the properties of the nucleons, making the proposed EIC the highest-resolution microscope of the nucleus. In summary, the EIC will facilitate an experimental setup that aims to answer the following outstanding questions [8]:

- How do the nucleonic properties such as mass and spin emerge from partons and their underlying interactions?
- How are partons inside the nucleon distributed in both momentum and position space?
- How do color-charged quarks and gluons, and jets, interact with a nuclear medium? How do the confined hadronic states emerge from these quarks and gluons? How do the quark-gluon interactions create nuclear binding?

1.4 The Focus of This Thesis

This thesis focuses on a computational study of electron–proton collisions using kinematics and beam parameters representative of the EIC. Events are generated with realistic Monte Carlo settings, then analyzed with modern jet-reconstruction algorithms to form experimentally relevant jets from final-state constituents. From each reconstructed jet, pertinent observables — transverse momentum p_t, pseudorapidity η, and azimuthal angle ϕ — are extracted. By studying the distributions of p_t, η, and ϕ across controlled kinematic bins, the analysis quantifies how parton-level information is reflected in detector-level jets and how reconstruction choices (clustering radius, constituent selection) bias those observables. The goal is twofold: (1) to validate the reconstruction chain and measure its resolution and systematic biases for EIC-like conditions, and (2) to identify which jet observables provide the most robust and sensitive probes of the underlying partonic dynamics in electron–proton collisions. Results and recommendations from these studies are intended to inform future experimental analyses and to provide a reproducible analysis workflow for subsequent EIC-related studies.

2 Computational Methodology

Due to the stochastic nature of quantum processes, the number of outgoing particles and their properties vary from event to event. The probability distributions for these properties can be inferred by studying an ensemble of events in data. Conversely, given a set of theoretically calculated probability distributions, it is possible to produce ensembles of simulated events to compare to data.

Numerical algorithm that generate random sequences of simulated events are called event generators. The simulations can be based on known or hypothetical laws of nature. This allows for the exploration and comparison of competing paradigms and studies of the sensitivity of proposed physical observables to the differences in experimental data [3].

Furthermore, to facilitate the computational simulation of the electron-proton collisions, this work utilizes PYTHIA, a Monte-Carlo event generator that provides the description of collisions at high energies between electrons, protons, photons, and heavy nuclei. Specifically, PYTHIA 8.3 is a scientific code library that is widely used for the generation of events in high-energy collisions between particles, where effects of the strong nuclear force, governed by QCD, are of high importance. It is written mainly in C++ and interweaves a comprehensive set of detailed physics models for the evolution from a few-body hard-scattering process to a complex multi-particle final state [3].

Moreover, in order to characterize the jets created in events generated by PYTHIA, this work utilizes FastJet, a C++ library that provides straightforward, efficient implementations for the majority of widely used jet recombination algorithms [5]. Jet recombination algorithms are systematic methods to project particles generated in collisions onto particular jets. There are two primary methods of recombination: sequential and cone. Sequential recombination algorithms typically identify the pair of particles that are closest in some distance measure, recombine them, and then continue iteratively until a stopping criterion is reached. Cone recombination algorithms group particles within specific conical angular regions, notably such that the momentum sum of the particles contained in a given cone coincides with the cone axis. In particular, this work utilizes the anti-k$_t$ algorithm, a sequential recombination algorithm provided by FastJet.

2.1 The Anti-k$_t$ Jet Algorithm

To understand the anti-k$_t$ algorithm, first consider the longitudinally-invariant k$_t$ algorithm, the first jet algorithm implemented in FastJet as well as a foundational source for other sequential recombination algorithms. As mentioned in the beginning of the *Computational Methodology* section, a sequential recombination algorithm characterizes jet behavior based on a chosen distance measure. Consider the symmetric distance measure:

$$d_{ij} = d_{ji} = min(p_{t,i}^2, p_{t,j}^2)\frac{\Delta R_{ij}^2}{R^2} \tag{10}$$

where $p_{t,i}$ represents the transverse momentum of the i-th particle with respect to the beam axis, which, by convention is the z-axis, $\Delta R_{ij}^2 = (\eta_i^2 - \eta_j^2)^2 + (\phi_i^2 - \phi_j^2)^2$, where ϕ_i is the azimuthal angle, and η_i is the pseudorapidity of the i-th particle, as defined in § 1.2.2. R is known as the *jet radius parameter*, a quantity, $0 < R \leq 1$, that determines the angular coverage of the algorithm. This algorithm also incorporates the distance of the i-th particle from the beam axis:

$$d_{iB} = p_{t,i}^2 \tag{11}$$

Based on these distance measures, jets are compartmentalized as follows:

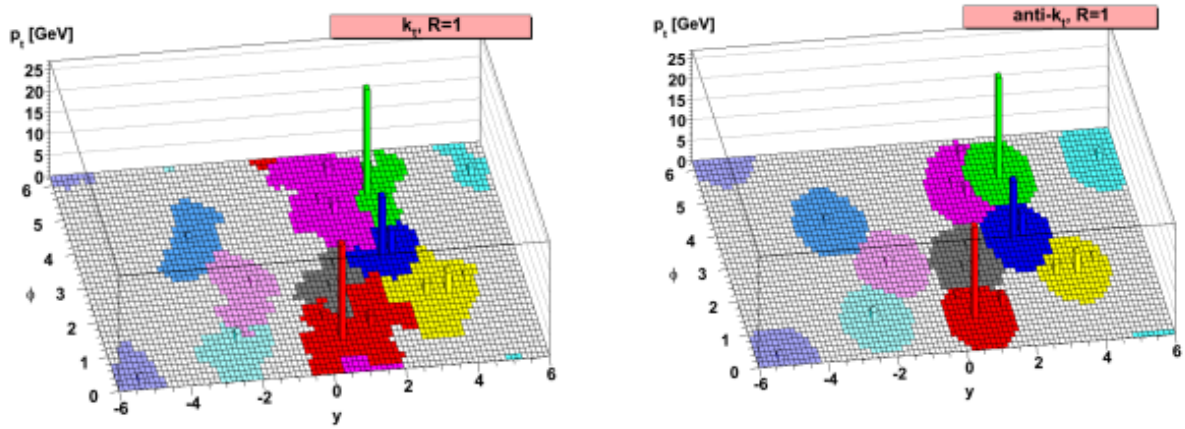
The algorithm identifies the smallest of d_{ij} and d_{iB}. If $d_{ij} < d_{iB}$, the objects i and j are replaced with a single object k, with momentum $p_k = p_i + p_j$. This object is classified as a "pseudojet". Conversely, if $d_B < d_{ij}$, then the i-th particle is omitted from the classification of jets created by the event. This procedure then iterates until the smallest d_{ij} or d_{iB} is above a chosen convergence threshold. The particles/pseudojets that remain are considered to be the jets of interest [5].

The anti-k$_t$ algorithm proceeds the same except it utilizes the reciprocal square of the momenta for the distance measure. That is:

$$d_{ij} = d_{ji} = min(\frac{1}{p_{t,i}^2}, \frac{1}{p_{t,j}^2})\frac{\Delta R_{ij}^2}{R^2} \tag{12}$$

$$d_{iB} = \frac{1}{p_{t,i}^2} \tag{13}$$

Consequently, the algorithm compartmentalizes jets with $y - \phi$ uniformity [4]. This effectively minimizes background "noise" from soft interactions, leaving only hard-scattered jets, exemplified in Figure 6.



(a) The k_t algorithm

(b) The anti-k_t algorithm

Figure 6: A visualization for the k_t and anti-k_t jet algorithms [4]. Note the conical uniformity in the $y - \phi$ plane for the anti-k_t algorithm.

Figure 6 provides a visualization for the k_t and anti-k_t jet clustering algorithms. The colored patches in the $y - \phi$ plane represent the jet's area/catchment region — i.e., the set of particles the jet algorithm assigned to that jet. The vertical colored spikes demonstrate the jet position, and the spike height is the jet p_t. Notice that in Figure 6 (b), the colored patches are nearly circular, showing an important property of the anti-k_t algorithm, in which jets are grouped in these uniform $y - \phi$ regions.

3 Procedure

Using PYTHIA, this thesis simulates the collision of 275 GeV protons with 18 GeV electrons, resulting in center-of-mass energies of approximately 140 GeV, which aligns with the anticipated experimental parameters for the EIC [12]. After simulating 10^6 collision events, we examine the kinematic variables of the final state particles. After validating these distributions, the anti-k_t jet clustering algorithm is applied to the events. We select jets with $p_t > 10$ GeV/c to investigate hard-scattered jet events, specifically. After examining the kinematics and multiplicities of these jets, i.e., the total number of jets created and the number per event, we tag them as either quark- or gluon-initiated. This thesis then examines the application of a generalized angularity function to distinguish between quark and gluon jets.

4 Analysis

4.1 Final State Particle Distributions

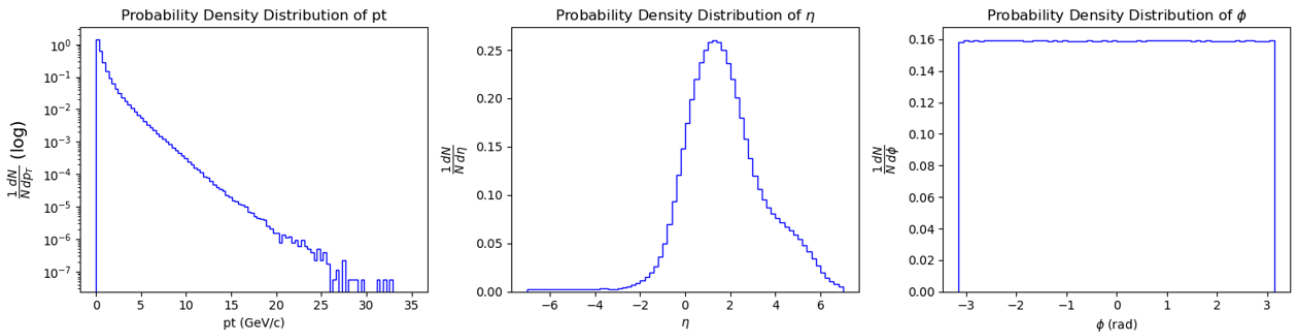


Figure 7: The final state charged particle kinematic distributions for 10^6 ep collisions. Distributions are normalized with respect to the total number of events such that the presented distributions are probability densities.

Figure 7 presents the final-state charged particle distributions corresponding to events generated by PYTHIA. Thus, these plots are representative of the final-state particles, without jet clustering. The plot on the left demonstrates the probability density distribution of the transverse momentum, plotted logarithmically. The sharp p_t fall-off illustrates the fact that particle production is dominated by soft (low p_t) processes. The middle plot is of the normalized, η distribution. One can see a notable asymmetry in η: in particular, a forward bias toward positive values. This is manifestly a result of our asymmetric, initial collisions. The proton is carrying significantly more momentum than the electron; thus, post-collision particles will be biased towards the direction of the proton's momentum. A qualitative depiction of this can be seen in Figure 8. The figure on the far right is the normalized ϕ distribution. Most notably, ϕ has a characteristic uniformity, which is a result of the fact that, based on the beam parameters, particles have no "preferred" direction of scattering in the azimuthal plane.

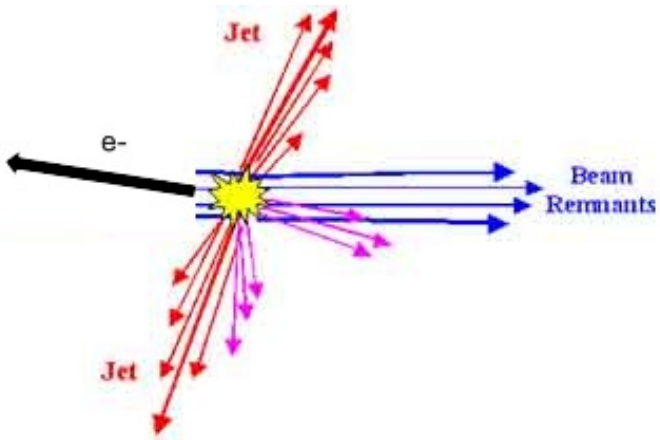


Figure 8: A qualitative depiction demonstrating the asymmetric ep collision. On the right-hand side, one can see how particle momenta and beam remnants are biased toward the initial direction of the proton.

4.2 Jet Multiplicities

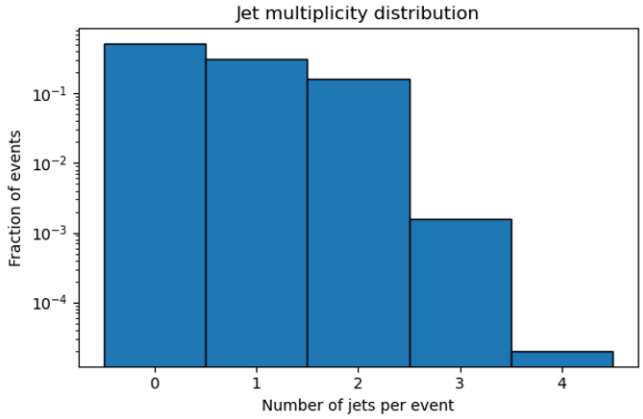


Figure 9: A histogram showing the multiplicities of jets with $p_t > 10$ GeV/c, expressed as a fraction of the total number of collision events. Of the 10^6 events, 645,336 produced jets with $p_t > 10$ GeV/c.

Figure 9 demonstrates the jet multiplicity distribution after applying the anti-k$_t$ clustering algorithm with a selection criterion of $p_t > 10$ GeV/c. Of the 10^6 events, 645,336 jets are created that satisfy this criterion. Additionally, 322,210 are dijet events, which we will use to perform an analysis to verify the kinematic distributions of our jets further.

4.3 Jet Kinematics

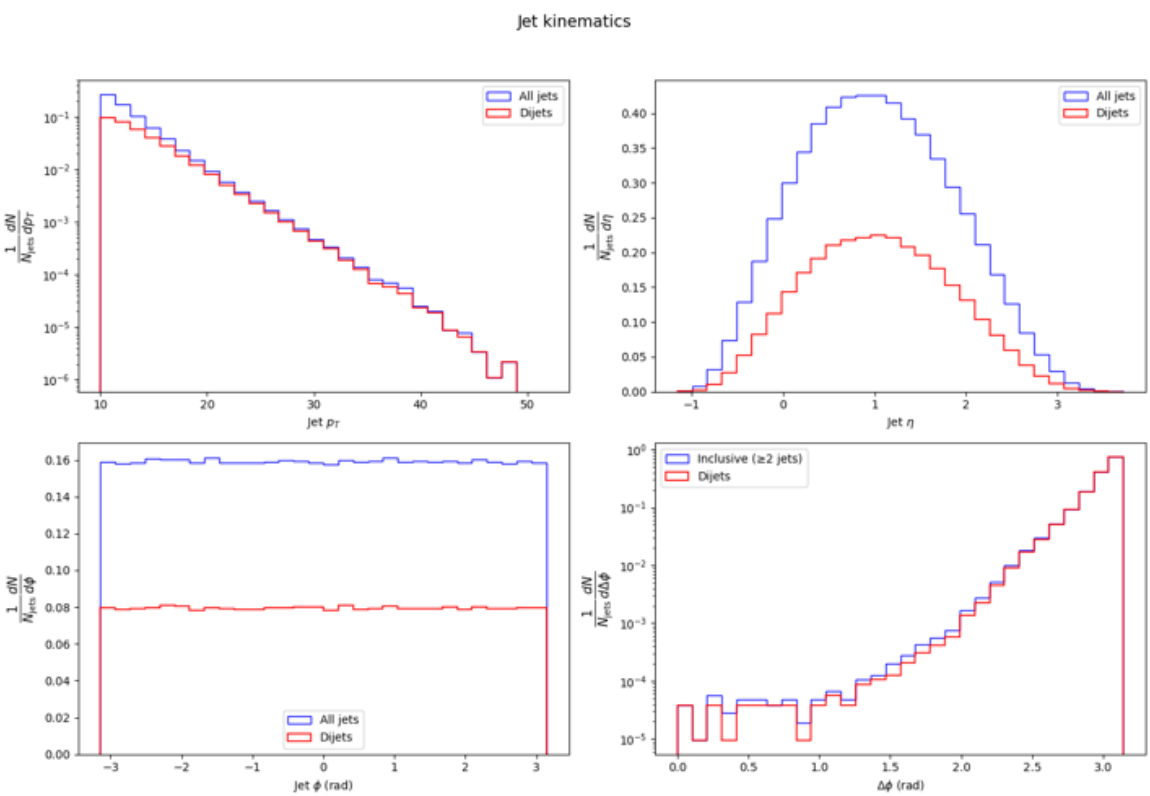


Figure 10: Four plots demonstrating the kinematic distributions of our jets, namely, p_t, η, ϕ, and $\Delta\phi$. The plots are normalized with respect to the total number of jets, such that the histograms represent probability densities. In the bottom right plot, for events with greater than two jets, the angular difference is calculated between the two highest transverse momentum jets.

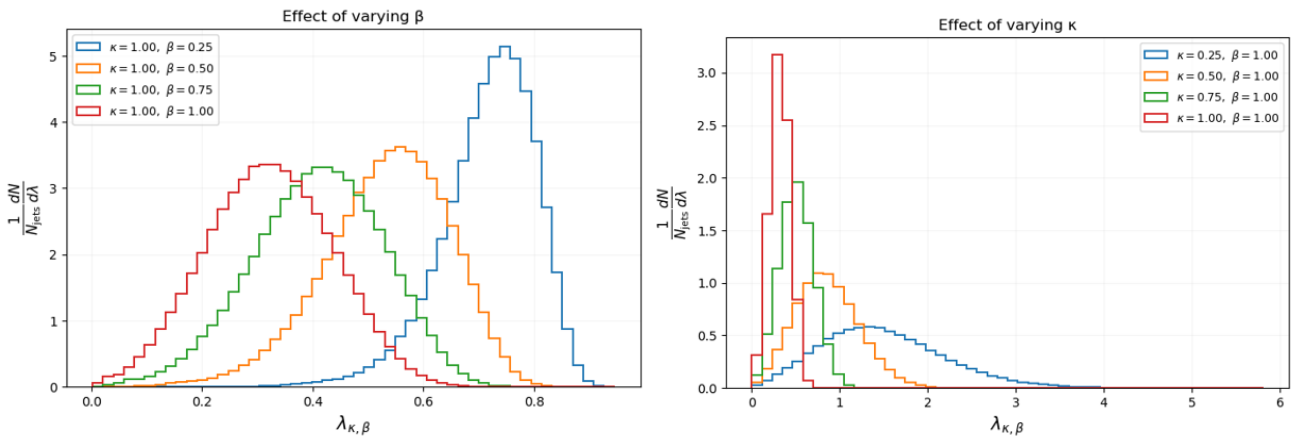
Figure 10 provides four plots, three of which are the standard p_t, η, and ϕ distributions, with the fourth being $\Delta\phi$, which represents the angular difference between two scattered jets. In the top left, we see the standard sharp p_t falloff. Note how dijet events dominate the high p_t regime, indicating that they largely occur via hard-scattering processes. The η and ϕ plots exhibit similar behavior as in Figure 7, that is, forward bias in η and symmetry in ϕ as we'd expect. The bottom right plot represents the azimuthal angular difference between the two leading transverse momentum jets for events with two or greater jets, plotted logarithmically. We can see that the overwhelming majority of angular differences are π radians, a manifestation of back-to-back jet production (see Figure 8), which can be attributed to momentum conservation.

4.4 Generalized Angularity Functions

As we have previously discussed, because we are relying on jets to probe our system, which are, by definition, reconstructed in a user-defined way, we introduce intrinsic biases that limit our ability to determine information about initial parton interactions with systematic certainty. Ultimately, the overarching motivation of studies similar to this thesis is to develop a framework in which we can recover observables that are relatively insensitive to noise and biases, such that we have a robust method to compare how these observables change under different initial conditions; for example, different initial beam energies or particle species. One such approach is through generalized angularity functions (GAFs). The particular GAF that this thesis utilizes is given as follows:

$$\lambda_\beta^\kappa = \sum_{i \in jet} z_i^\kappa \left(\frac{\Delta R_i}{R}\right)^\beta \tag{14}$$

where $z_i = p_{t,c}/p_{t,jet}$ is the fractional momentum carried by the i-th jet constituent, ΔR_i and R are as defined in § 2.1, and κ and β are user-defined weights that scale the momentum and angular contributions of the jets, respectively. GAFs can be used to characterize jet shapes and have previously been used to discriminate between quark and gluon jets, which is the manner in which they will be employed for this thesis [6]. After evaluating and demonstrating the effects of varying κ for a fixed β value, and vice versa, this thesis investigates the efficacy of utilizing GAFs to distinguish quark and gluon jets across several weights, in order to determine if a particular class of user-defined weights is best for this application.

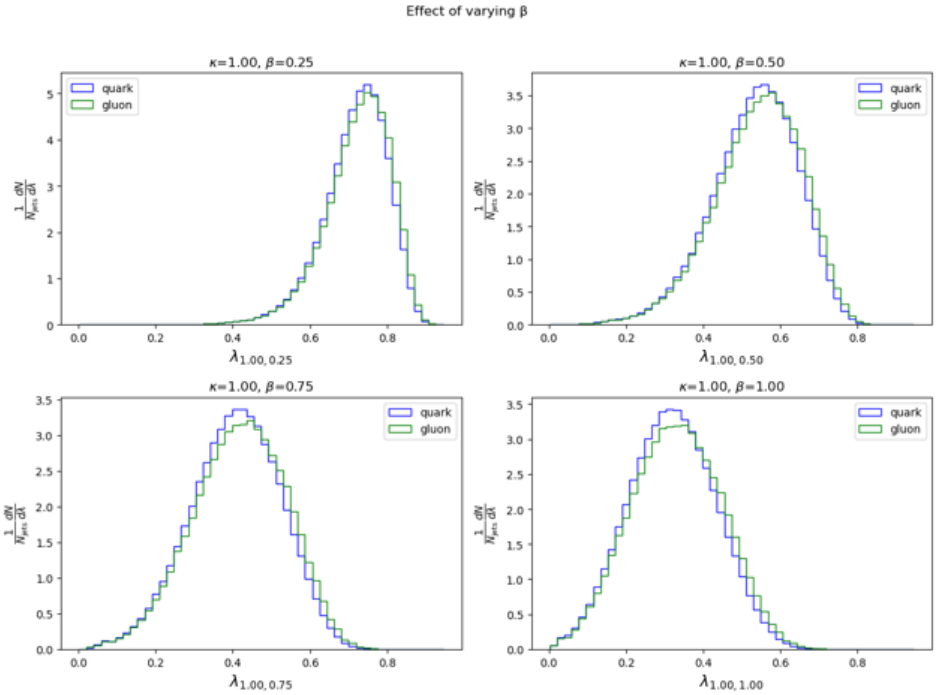


(a) The effect of varying β

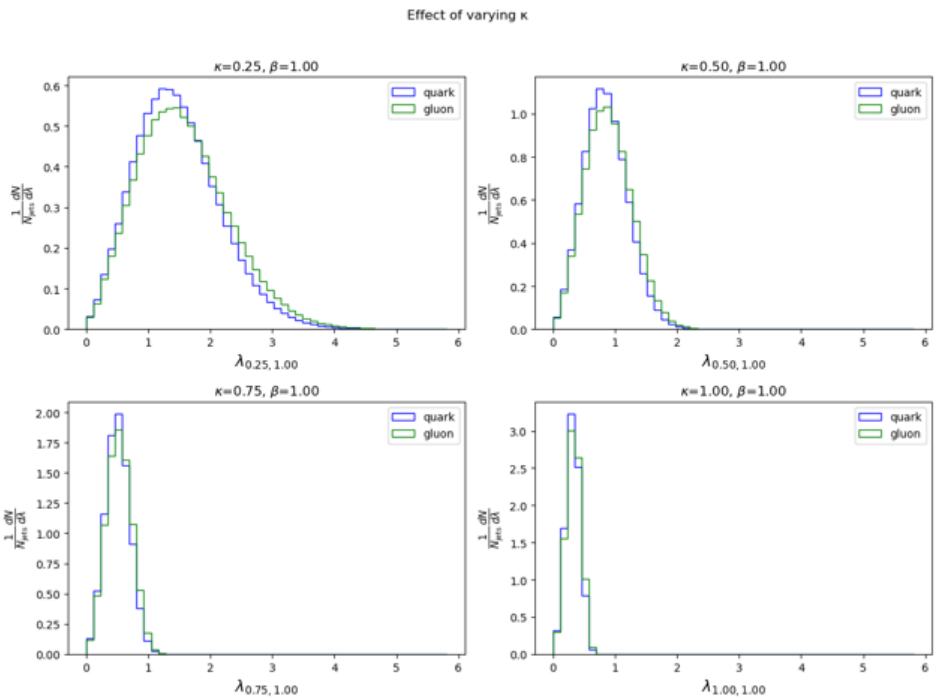
(b) The effect of varying κ

Figure 11: The application of the generalized angularity function to the jets created in *ep* collisions. Figure (a) demonstrates the effect of varying β from .25 to 1 in increments of .25 for κ fixed at 1. Conversely, Figure (b) demonstrates the effect of varying κ from .25 to 1 increments of .25 while keeping β fixed at 1.

Figure 11 demonstrates the effect of varying weights on the chosen GAF. Figure 11 (a) shows the behavior of the GAF for a variable β but fixed $\kappa = 1$. Manifestly, increasing β shifts the GAF distribution toward lower values, while broadening its peak. Similarly, increasing κ with β fixed shifts the distribution toward lower values; however, this results in a considerable sharpening of the peak. We use this information to inform expected distributions for quark- and gluon-tagged jets.



(a) The effect of varying β on quark/gluon GAF distributions



(b) The effect of varying κ on quark/gluon GAF distributions

Figure 12: The application of the generalized angularity function to quark (blue) and gluon (green) jets. Figure (a) demonstrates the effect of varying β from .25 to 1 in increments of .25 for κ fixed at 1. Conversely, Figure (b) demonstrates the effect of varying κ from .25 to 1 increments of .25 while keeping β fixed at 1.

Figure 12 demonstrates the application of the GAF to quark and gluon jets. Overall, varying the weights modifies the distributions similarly to Figure 11. However, because the GAF provides insight into quark and gluon jet shapes, we ultimately implement this procedure to investigate differences between the quark and gluon GAF distributions as a proxy for their physical shapes. According to QCD, gluons carry a bigger color charge than quarks. Therefore, they radiate more soft/collinear QCD radiation, meaning gluon-initiated showers are denser and spread out to larger angles. That extra radiation makes gluon jets broader, with higher particle multiplicity and softer fragments than quark jets [17]. Because the GAF can be used as a proxy for jet shape, we should expect this physical difference to manifest in the distributions. By inspection, the gluonic distributions are perhaps shifted slightly toward higher values, but it is difficult to make any definitive conclusions regarding any differences between the quark and gluon distributions. An important factor to consider in this analysis is that we utilize a jet radius parameter, $R = .4$, typical for studies in heavy-ion collisions, where large angular coverage can result in contamination of desired data. Nonetheless, imposing this angular coverage omits wide-angle jets, akin to those created by gluons. Therefore, this analysis is likely to only accept gluon jets with limited angular broadening; thus, the quark and gluon jets we collect possess a similar shape, and consequently, their GAF distributions appear to be similar. As discussed in § 1.3, in contrast to heavy-ion collisions, where there is considerable "noise", *ep* collisions offer a clean experimental probe of partonic physics; ergo, increasing the angular coverage is not likely to contaminate any signal-to-noise ratio, and can potentially reveal the full behavior of the gluon jet shapes, which would manifest in their GAF distributions.

5 Conclusions and Future Outlook

In summary, this thesis presents a simulation study of *ep* collisions at $\sqrt{s} = 140$ GeV, energies representative of the future Electron-Ion Collider. Using PYTHIA, we simulate 10^6 events, and extract p_t, η, and ϕ of the final-state particles. After validating these kinematics, we apply the anti-k_t jet clustering algorithm. We conduct a similar procedure to validate the kinematics. We then apply a generalized angularity function to all jets created. We evaluate the effect that varying the weights has on the distributions, before applying the GAF to quark- and gluon-tagged jets. By systematically varying the weights, we seek discrepancies between quark and gluon jet shapes. While the gluonic distributions were minutely shifted toward higher values, we cannot make any definitive conclusions regarding our ability to discriminate between quark and gluon jets using the generalized angularity function.

The natural continuation for this study would be to analyze the generalized angularity functions across various collision systems to be investigated at the EIC, such as eAu and ePb, and to systematically investigate differences between these systems. This can be done through the Benchmark eA Generator for Leptoproduction (BeAGLE) software, which provides the source code necessary to facilitate electron-nucleus Monte Carlo simulations [10].

6 Acknowledgments

I would like to thank Ravi Koka, Khovesh Ramdin, Tanmay Pani and Professor Sevil Salur, whose mentorship was indispensable in the making of this thesis.

References

[1] Abdul Khalek et al. "Science Requirements and Detector Concepts for the Electron-Ion Collider". In: *Nuclear Physics A* 1026 (Oct. 2022), p. 122447. ISSN: 0375-9474. DOI: 10.1016/j.nuclphysa.2022.122447. URL: http://dx.doi.org/10.1016/j.nuclphysa.2022.122447.

[2] Siegfried Bethke. "Experimental tests of asymptotic freedom". In: *Nuclear Physics B - Proceedings Supplements* 58 (Apr. 2007), pp. 314–324. DOI: 10.1016/S0920-5632(97)00061-3.

[3] Christian Bierlich et al. *A comprehensive guide to the physics and usage of PYTHIA 8.3*. 2022. arXiv: 2203.11601 [hep-ph]. URL: https://arxiv.org/abs/2203.11601.

[4] Matteo Cacciari, Gavin P Salam, and Gregory Soyez. "The anti-ktjet clustering algorithm". In: *Journal of High Energy Physics* 2008.04 (Apr. 2008), pp. 063–063. ISSN: 1029-8479. DOI: 10.1088/1126-6708/2008/04/063. URL: http://dx.doi.org/10.1088/1126-6708/2008/04/063.

[5] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. "FastJet user manual: (for version 3.0.2)". In: *The European Physical Journal C* 72.3 (Mar. 2012). ISSN: 1434-6052. DOI: 10.1140/epjc/s10052-012-1896-2. URL: http://dx.doi.org/10.1140/epjc/s10052-012-1896-2.

[6] CMS Collaboration. "Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions". In: *JHEP* 01 (2022). CERN-EP-2021-161, CMS-SMP-20-010, p. 188. DOI: 10.1007/JHEP01(2022)188. arXiv: 2109.03340 [hep-ex].

[7] H1 Collaboration et al. *Measurement of lepton-jet correlation in deep-inelastic scattering with the H1 detector using machine learning for unfolding*. 2022. arXiv: 2108.12376 [hep-ex]. URL: https://arxiv.org/abs/2108.12376.

[8] EIC User Group. *White Paper on the Electron–Ion Collider in Preparation for the NSAC Long Range Plan*. Zenodo. Accessed: 2025-10-22. Jan. 2023. DOI: 10.5281/zenodo.7500024. URL: https://doi.org/10.5281/zenodo.7500024.

[9] Dave Gaskell. "Electron Polarimetry". In: Sept. 2022, pp. 301–314. ISBN: 978-3-031-16714-0. DOI: 10.1007/978-3-031-16715-7_13.

[10] GitHub. *eic*. https://github.com/eic. GitHub repository, accessed 2025-MM-DD. 2025.

[11] Ravi Koka. *Acceptance Studies for ALICE at the Large Hadron Collider: Analyzing the Effect of Detector Acceptance on Hadron–Λ Angular Correlations, as Part of an Investigation of the Quark–Gluon Plasma in Small Systems*. Undergraduate thesis, University of Texas at Austin. Advisor: Christina Markert; Department of Physics. May 2024. URL: https://github.com/ravikoka/broad/tree/main/thesis.

[12] D Marx et al. "Designing the EIC electron storage ring lattice for a wide energy range". In: *Journal of Physics: Conference Series* 2420.1 (Jan. 2023), p. 012010. DOI: 10.1088/1742-6596/2420/1/012010. URL: https://doi.org/10.1088/1742-6596/2420/1/012010.

[13] E. Rutherford. "The Scattering of
alpha and
beta Particles by Matter and the Structure of the Atom". In: *Philosophical Magazine*. 6th ser. 21.May (1911), pp. 669–688.

[14] Richard E. Taylor. "The Discovery of the point - like structure of matter". In: *Phil. Trans. Roy. Soc. Lond. A* 359 (2001). Ed. by I. Butterworth, John R. Ellis, and E. Gabathuler, pp. 225–240. DOI: 10.1098/rsta.2000.0723.

[15] TikZ.net contributors. *axis3d_cms — TikZ example*. TikZ.net (online). Accessed: 2025-10-22. 2025. URL: https://tikz.net/axis3d_cms/.

[16] Wikimedia Commons contributors. *File:Standard Model of Elementary Particles.svg*. Wikimedia Commons (image). Accessed: 2025-10-20. URL: https://en.wikipedia.org/wiki/File:Standard_Model_of_Elementary_Particles.svg.

[17] W. Zeuner and OPAL Collaboration. "Differences between jets from quarks and gluons". In: *29th Rencontres de Moriond — QCD and High Energy Hadronic Interactions (Hadronic Session)*. CERN record 295495. 1994, pp. 125–130. URL: https://cds.cern.ch/record/295495/files/C94-03-19_125-130.pdf.